



Microbiota-derived short-chain fatty acids mediate the effects of dengzhan shengmai in ameliorating cerebral ischemia via the gut–brain axis

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ABSTRACT

Ethnopharmacological relevance: Dengzhan shengmai (DZSM) formula, composed of four herbal medicines (*Erigeron breviscapus*, *Panax ginseng*, *Schisandra chinensis*, and *Ophiopogon japonicus*), is widely used in the recovery period of ischemic cerebrovascular diseases; however, the associated molecular mechanism remains unclear.

Aim of the study: The purpose of this study was to uncover the links between the microbiota–gut–brain axis and the efficacy of DZSM in ameliorating cerebral ischemic diseases.

Materials and methods: The effects of DZSM on the gut microbiota community and bacteria-derived short-chain fatty acid (SCFA) production were evaluated *in vivo* using a rat model of cerebral ischemia and *in vitro* through the anaerobic incubation with fresh feces derived from model animals. Subsequently, the mechanism underlying the role of SCFAs in the DZSM-mediated treatment of cerebral ischemia was explored.

Results: We found that DZSM treatment significantly altered the composition of the gut microbiota and markedly enhanced SCFA production. The consequent increase in SCFA levels led to the upregulation of the expression of monocarboxylate transporters and facilitated the transportation of intestinal SCFAs into the brain, thereby inhibiting the apoptosis of neurocytes via the regulation of the PI3K/AKT/caspase-3 pathway. The increased intestinal SCFA levels also contributed to the repair of the 2VO-induced disruption of gut barrier integrity and inhibited the translocation of lipopolysaccharide from the intestine to the brain, thus attenuating neuro-inflammation. Consequently, cerebral neuropathy and oxidative stress were significantly improved in 2VO model rats, leading to the amelioration of cerebral ischemia-induced cognitive dysfunction. Finally, fecal microbiota transplantation could reproduce the beneficial effects of DZSM on SCFA production and cerebral ischemia.

Conclusions: Our findings suggested that SCFAs mediate the effects of DZSM in ameliorating cerebral ischemia via the gut microbiota–gut–brain axis.

1. Introduction

Cerebral ischemia, caused by focal vascular occlusion or arterial stenosis, is a major risk factor for several neurodegenerative conditions, including vascular dementia and Alzheimer's disease (Sheng et al.,

2020). Ischemic cerebrovascular events account for 60%–80% of all cerebrovascular diseases and can result in the obstruction of local blood circulation, necrosis of brain tissue, cerebral edema, neurological impairment, and cognitive defects (Liu et al., 2020). Cerebral ischemic stroke is the second leading cause of death and the third most common cause of disability worldwide (Pluta et al., 2021). However, safe and

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Abbreviations:

DZSM	dengzhan shengmai
SCFA	short-chain fatty acid
LPS	lipopolysaccharide;
TCM	traditional Chinese medicine;
SOD	superoxide dismutase
MDA	malondialdehyde
DOX	doxorubicin
Na-AA	sodium acetate
Na-PA	sodium propionate
Na-BA	sodium butyrate
2VO	permanent occlusion of bilateral common carotid arteries
FABP2	fatty acid binding protein 2
MWM	morris water maze
AB-PAS	alcian blue periodic acid-Schiff
MCT1	monocarboxylate transporter 1
SMCT1	sodium-coupled monocarboxylate transporter 1
OCLN	occludin

effective therapies are still lacking owing to the complex pathogenesis of the condition, which involves mechanisms such as energy metabolism impairment, metabolic acidosis, calcium overload, excitotoxicity, free radical production-induced oxidative stress, neuronal apoptosis, cerebral inflammation, and autophagy (Chen et al., 2019b). Accordingly, novel therapeutic options for the treatment of cerebral ischemia are urgently required.

The gut–brain axis comprises a bidirectional communication system via which gut microbes interact with the brain (Socala et al., 2021). Clinical trials have demonstrated that cerebral ischemic diseases are often accompanied by gastrointestinal symptoms and gut dysbiosis (Zeng et al., 2019). Increasing evidence supports that the gut microbiota modulates the progression and prognosis of cerebral ischemic stroke (Pluta et al., 2021). Additionally, cerebral ischemia can disturb the composition of the microbiota and the contents of bacteria-derived metabolites in the gut, as well as disrupt gut barrier integrity. Patients or Sprague–Dawley (SD) rats that have experienced cerebral ischemic stroke harbor a greater number of opportunistic pathogens, such as *Alistipes*, *Klebsiella*, *Shuttleworthia*, *Haemophilus*, *Fusobacterium*, *Proteus*, and *Papillibacter*, but fewer beneficial types, such as *Lactobacillus*, *Butyrivibrio*, *Meganon*, *Prevotella*, and *Alloprevotella* (Chen et al., 2019b; Yin et al., 2015). The production of several important gut bacteria-derived metabolites, such as short-chain fatty acids (SCFAs), microbial amyloid peptides, and trimethylamine, has been associated with treatment outcomes in patients with ischemic brain injury (Pluta et al., 2021). SCFAs are particularly important owing to their wide-ranging physiological functions, including the regulation of cerebral neurotransmitters, microglia-mediated immunoinflammatory responses, and apoptosis of cerebral neurocytes (Silva et al., 2020). Increased gut permeability is considered a key risk factor for cerebral ischemic stroke because a leaky gut can lead to the release of enterotoxins, such as lipopolysaccharide (LPS), into the blood circulation and brain tissues, which activates immunoinflammatory responses and results in cerebral neuronal necrosis (Chen et al., 2019a). SCFAs also play an important role in maintaining intestinal barrier homeostasis by inhibiting the growth of opportunistic pathogens, improving the intestinal mucus barrier, and modulating the expression of several tight junction proteins in enterocytes (Chen et al., 2019b). Thus, targeting SCFA production pathways may hold promise as a therapeutic strategy for cerebral ischemic diseases.

Traditional Chinese medicine (TCM) represents a rich source of candidate drugs for the treatment of cerebrovascular diseases. Dengzhan

shengmai (DZSM) formula, composed of four herbal medicines—the herba of *Erigeron breviscapus* (Vaniot) Hand.-Mazz, the root of *Panax ginseng* C.A.Mey, the fruit of *Schisandra chinensis* (Turcz.) Baill; and the root of *Ophiopogon japonicus* (Thunb.) Ker Gawl—has been shown to be safe and effective in the treatment of ischemic stroke and dementia based on systematic reviews and a meta-analysis of randomized controlled trials (Cai et al., 2019; Liu et al., 2022). The plant names of all herbal medicines have been checked at <http://www.theplantlist.org>. It was approved by the National Medical Products Administration of China in 2007 and is listed in the National Essential Drugs List of China (Sheng et al., 2020). Besides, some plants used in the formulation of DZSM are not unique to TCM and are widely used in other systems of medicine, suggestive of their effectiveness and safety as herbal medicines. For example, *Panax ginseng* and *Schisandra chinensis* are also used in the official medicine of many countries and is included in the European, Japanese, Russian, and American Herbal Pharmacopoeias (European Pharmacopoeia Committee, 2019; Japanese Pharmacopoeia Committee, 2021; Panossian et al., 2021; Upton et al., 2011). Meanwhile, *Ophiopogon japonicus* is included in the Japanese Pharmacopoeia (Japanese Pharmacopoeia Committee, 2021). Additionally, *Schisandra chinensis* and *Panax ginseng* are well-established adaptogens used in Russia and have been demonstrated to possess pleiotropic pharmacological effects (Panossian et al., 2021). Each plant that comprises DZSM has a specific therapeutic mechanism against cerebral ischemia. *Erigeron breviscapus* promotes the dilation of arterioles, improves microcirculation, reduces platelet aggregation, and retards thrombosis, thereby exerting protective effects on the vascular endothelium (Wu et al., 2021). *Panax ginseng* promotes the proliferation and differentiation of neurocytes, inhibits amino acid excitotoxicity and intracellular calcium overload, and contributes to the maintenance of mitochondria function and blood–brain barrier integrity (Yang, F. et al., 2020a). *Schisandra chinensis* exerts neuroprotective effects via the regulation of oxidative stress, apoptosis, inflammation, neurotransmitter release, and brain-derived neurotrophic factors (Zhang et al., 2018). Finally, *Ophiopogon japonicus* exhibits anti-ischemic effects mainly through mechanisms that include anti-oxidation, correcting acidosis, promoting astrocyte proliferation, and delaying brain aging (Gao et al., 2021). Indeed, all four plants are essential for the effects of DZSM as its efficacy is dependent on the synergistic effects of its multiple bioactive constituents. However, owing to the complex composition of DZSM, the molecular mechanisms underlying its effects have remained unclear. Although several active ingredients, such as scutellarin, 4,5-dicaffeoylquinic acid, 3, 5-dicaffeoylquinic acid, ginsenoside Rb1, and schizandrin A, have been reported to be associated with the effects of DZSM in the treatment of cerebral ischemia (Liu et al., 2022; Sheng et al., 2020; Xie et al., 2021; Zhou et al., 2019), these compounds cannot be absorbed into the blood circulation in large amounts due to poor solubility in water, large molecular mass, or poor enterocyte permeating ability, and are thus retained in the intestine (Bi et al., 2021; Chen et al., 2020), where they may directly interact with the gut microbiota. Accordingly, investigating the role of the gut microbiota in the therapeutic effects of DZSM against cerebral ischemia may improve our understanding of its mechanism of action.

In this study, we aimed to uncover the mechanism underlying DZSM's anti-cerebral ischemic effects through gut–brain crosstalk. We investigated how DZSM affects the gut microbiota and bacteria-mediated SCFA production and transportation *in vivo*, using a rat model of cerebral ischemia, and *in vitro*, through anaerobic incubation with fresh feces collected from model animals. Furthermore, the contribution of SCFAs to the DZSM-induced amelioration of cerebral ischemia was explored from the perspective of neuronal apoptosis and neuroinflammation.

2. Materials and methods

2.1. Materials

The four TCMs *Erigeron breviscapus* (Lot: 181101), *Radix ginseng* (Lot: 190205), *Schisandra chinensis* (Lot: 190107), and *Radix ophiopogonis* (Lot: 180403) were purchased from Sanyue Chinese Traditional Medicine Co., Ltd (Nantong, China). Each herbal medicine was identified by the company and passed quality inspection (reports were provided). Caco-2 cells and PC12 cells were obtained from the Cell Resource Center of Peking Union Medical College (Beijing, China). Superoxide dismutase (SOD) and malondialdehyde (MDA) assay kits were obtained from the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). The ELISA kits used for the determination of LPS, fatty acid-binding protein 2 (FABP2), zonulin, IL-1 β , IL-6, and TNF- α levels were bought from Enzyme-linked Biotechnology Co., Ltd (Shanghai, China). Trypsin-EDTA (0.25%), cell culture medium, penicillin/streptomycin, and fetal calf serum (FBS) were purchased from Thermo Fisher Scientific (Waltham, MA, USA). The standards for the DZSM ingredients were purchased from Yuanye Biotechnology Co., Ltd (Shanghai, China). The analytical standards of volatile free acid mix, doxorubicin (DOX), LPS, sodium acetate (Na-AA), sodium propionate (Na-PA), and sodium butyrate (Na-BA) were purchased from Sigma-Aldrich (St Louis, MO, USA). All other reagents were of analytical or HPLC grade.

2.2. Preparation and chemical profiles of DZSM

The extraction and preparation of DZSM were undertaken by Biovalley Pharmaceutical Co., Ltd (Kunming, China; Batch number: 20190023) based on the China Pharmacopoeia ([China Pharmacopoeia Committee, 2020](#)). Briefly, the medicinal materials of DZSM powder were weighed in the proportions of *Erigeron breviscapus*: *Radix ginseng*: *Schisandra chinensis*: *Radix ophiopogonis* = 5:1:1:1.8. First, all the medicinal materials were extracted three times with 90% ethanol and concentrated by reflux. Then, the ethanol extract of ginseng, *Schisandra chinensis*, and *Ophiopogon* was further extracted three times using *n*-butyl alcohol and concentrated *in vacuo*. Finally, the two extracts were combined in distilled water and spray-dried, yielding DZSM powder. The chemical profiles of DZSM were determined using a 1290 Infinity II ultra-high-performance liquid chromatography system coupled to a 6550 iFunnel quadrupole time-of-flight mass spectrometer equipped with a dual AJS electrospray ionization source (UHPLC–HRMS, Agilent, USA). The components of DZSM were separated on an Agilent Zorbax SB-C18 column (150 $\times$ 4.6 mm, 5 μ m particle size). Sample analysis in both negative and positive ionization modes was performed using water containing 0.1% formic acid as mobile phase A and acetonitrile as mobile phase B. The elution gradient used was as follows: isocratic step at 10%–20% B in 5 min, 20%–60% B in 20 min, 60%–95% B in 25 min, and 95%–100% B in 30 min; finally, the column was equilibrated at initial conditions for 5 min. The flow rate was 0.4 mL/min, the injection volume was 5 μ L, and the column oven temperature was maintained at 35 $^{\circ}$ C.

2.3. Cell culture and treatment

Caco-2 cells were grown in minimum Eagle's medium supplemented with 20% heat-inactivated FBS and 1% nonessential amino acids in an atmosphere with 5% CO₂ at 37 $^{\circ}$ C. Cells were incubated for 24 h with Na-AA (500 μ M), Na-PA (500 μ M), Na-BA (500 μ M), or DZSM (250 or 500 μ g/mL). Differentiated PC12 cells were grown in RPMI-1640 medium supplemented with 10% FBS. LPS (1 μ g/mL)- or DOX (0.5 μ M)-treated PC12 cells were incubated with Na-AA (50 μ M), Na-PA (25 μ M), Na-BA (25 μ M), or DZSM (50 or 100 μ g/mL) for 24 h. A CCK-8 Assay Kit (Beyotime Biotechnology, Shanghai, China) was used to test PC12 cell viability. For the apoptosis assay, DOX- or drug-treated PC12 cells were subjected to flow cytometry (Becton Dickinson, San Jose, CA, USA)

using an Annexin V-FITC/PI Apoptosis Detection Kit (Beyotime Biotechnology) according to the manufacturer's instructions. The proportion of apoptotic PC12 cells was determined using FlowJo software (TreeStar, Ashland, OR, USA).

2.4. Animals

SD rats (male, 6 weeks old, 180–200 g) were bought from Charles River Laboratory Animal Technology Co., Ltd (Beijing, China). The animals were housed in a specific-pathogen-free (SPF) environment under controlled conditions (temperature: 22 $\pm$ 2 $^{\circ}$ C; relative humidity: 50 $\pm$ 5%; and a 12/12 h light/dark cycle) and had free access to food and water. All studies involving animals were performed in compliance with institutional guidelines and ethics and were approved by the Laboratories Institutional Animal Care and Use Committee of the Chinese Academy of Medical Sciences and Peking Union Medical College (No. 00009051).

2.5. Generation of a rat model of permanent occlusion of bilateral common carotid arteries (2VO)

In this study, the 2VO model was utilized to induce permanent and persistent cerebral ischemia in SD rats. All the rats had access to water but not food for 12 h before 2VO surgery. Each rat was anesthetized with a mixture of 5% isoflurane and 95% oxygen. A mixture of 3% isoflurane and 97% oxygen was supplied via a respiratory mask during surgery. In the permanent 2VO group (n = 40), following a 1.5–2-cm midline cervical incision, both common carotid arteries were carefully exposed and doubly ligated with 5-0 silk sutures just below the bifurcation into the internal and external carotid arteries. The sham-operated animals (n = 10) were treated in the same manner as the 2VO rats but without carotid artery ligation. The wound was then sprinkled with an appropriate amount of penicillin powder and carefully closed.

2.6. The treatment of 2VO rats

Four weeks after surgery, the model rats were randomly divided into the following four groups: a model group (treated with distilled water, n = 9), a DZSM group [720 mg/(kg $\cdot$ day) DZSM, n = 9], a DZSM + antibiotics (D + AB) group [720 mg/(kg $\cdot$ day) DZSM + 200 mg/(kg $\cdot$ day) amoxicillin + 200 mg/(kg $\cdot$ day) clarithromycin + 200 mg/(kg $\cdot$ day) metronidazole, n = 9], and a SCFA group [200 mg/(kg $\cdot$ day) Na-AA + 100 mg/(kg $\cdot$ day) Na-PA + 100 mg/(kg $\cdot$ day) Na-BA, n = 10]. The sham operation group (n = 9) served as a control. All the rats were treated with distilled water, DZSM, antibiotics, or SCFAs by gavage (2 mL per rat) for 4 weeks. Moreover, to minimize drug–drug interaction in the D + AB group, antibiotics and DZSM were administered at different time points.

After treatment, the rats were euthanized with isoflurane and decapitated. Fresh colonic feces were collected, snap-frozen in liquid nitrogen, and stored at -80° C for further use. Blood samples were centrifuged (3,500 rpm, 10 min) for serum separation and all serum aliquots were stored at -80° C. Colon tissues were fixed for histological staining or stored at -80° C. Half of the rat brain was fixed for histological staining. The hippocampus and cortex were separated from the other half of the brain and kept at -80° C for further analysis. The levels of LPS, FABP2, zonulin-1, SOD, MDA, TNF- α , IL-1 β , and IL-6 in tissues or plasma were quantified using commercial kits following standard protocols.

2.7. Fecal Microbiota Transplantation (FMT)

Fresh feces (2 g) were collected daily from rats in the model and the DZSM groups, homogenized in sterile PBS, and centrifuged at 100 rpm for 1 min at 4 $^{\circ}$ C. The precipitates were then washed three times with PBS. All the supernatants were collected and centrifuged at 5,000 rpm

for 20 min and the resulting bacterial precipitates were suspended in PBS (20 mL) and administered to antibiotic-pretreated 2VO rats by gavage (2 mL per rat) for 4 weeks ($n = 8$ per group).

2.8. Morris water maze (MWM) test

The MWM test was performed before and after drug treatment. The maze consisted of a circular water tank 120 cm in diameter and 50 cm in height filled to a depth of 30 cm with water at $23 \pm 1^\circ\text{C}$ that covered a black platform 10 cm in diameter. The tank was divided into four equal quadrants. The top of the platform, located in the center of quadrant I, was approximately 1 cm below the surface of the water. The swimming path of each rat was monitored using a SuperMaze Morris image system (Softmaze, Shanghai, China). The behavioral experiments included an orientation navigation test that lasted for 3 days, performed three times per day, during which rats were allowed 90 s to find the hidden platform, and the escape latency was recorded; and a spatial exploration test (three times for 1 day), in which, after a 3-day training trial, the hidden platform was removed, and the rats were allowed to swim freely for 90 s to record the number of crossing the original platform location, the time spent in the target quadrant, and the time spent searching the original platform location.

2.9. *In vitro* anaerobic incubation of fresh feces

Fresh feces (2 g) were collected from 2VO rats and pre-incubated for 60 min in normal saline (feces: saline [w/v] = 1:20) under anaerobic conditions (N_2 : H_2 : $\text{CO}_2 = 8:1:1$, 37°C) and mixed thoroughly to obtain intestinal bacterial cultures. DZSM (final concentration: 250, 500, or 1000 $\mu\text{g/mL}$) or its various ingredients were added to the intestinal bacterial cultures followed by incubation for 18 h at 37°C . Distilled water served as a negative control. Simultaneously, an equal amount of intestinal bacteria was killed by boiling for 15 min. After incubation, bacterial cultures were centrifuged at 10,000 rpm for 10 min at 4°C and the supernatants were collected for further analysis.

2.10. Analysis of the gut microbiota

Total bacterial DNA was extracted from colonic contents using a QIAamp Fast DNA Stool Mini Kit (Qiagen, Frankfurt, Germany). The gut bacterial composition of the samples was analyzed via 16S rRNA gene sequencing. A sequencing library was constructed using the 341F/806R primer pair targeting the V3–V4 region and paired-end sequenced in the Illumina MiSeq platform. Sequences were assigned to operational taxonomic units (OTUs, 97% identity) after quality filtering, followed by the selection of representative sequences using the QIIME software package. The Majorbio I-Sanger Cloud Platform was used to analyze the microbiota (<https://cloud.majorbio.com/>). Alpha-diversity, beta-diversity, and bacterial community composition were analyzed using the QIIME tool. Linear discriminant analysis (LDA) effect size (LEfSe) was performed with an LDA score >2.0 .

2.11. RT-qPCR

Total RNA was extracted from cell precipitates, colonic contents, colon, and hippocampal tissues using the TRIzol Plus RNA Purification Kit (Invitrogen, CA, USA). RT-qPCR was performed on an ABI 7500 Fast System (Applied Biosystems, CA, USA) using the Power SYBR Green RNA-to-CT 1-Step Kit (Applied Biosystems). GAPDH was used for normalization. The results are presented as fold changes relative to GAPDH levels and were calculated using the $2^{-\Delta\Delta\text{CT}}$ method. The sequences of the primers used for qPCR are shown in Table S1.

2.12. Western blot

Total protein was extracted in protein lysis buffer premixed with

protease/phosphatase inhibitors (Thermo Fisher Scientific). Equal amounts of protein were then resolved by SDS-PAGE, transferred to polyvinylidene difluoride (PVDF) membranes (Millipore, Billerica, MA, USA), blocked with 5% non-fat milk powder in TBST at room temperature for 2 h, and incubated with primary antibodies targeting monocarboxylate transporter 1 (MCT1, Lot: ER1912-87, Huabio, Hangzhou, China); sodium-coupled monocarboxylate transporter 1 (SMCT1, Lot: 21433-1-AP), ZO-1 (21773-1-AP), occludin (OCLN, Lot: 27260-1-AP), and GAPDH (Lot: 10494-1-AP, Proteintech, Chicago, IL, USA); and AKT (Lot: #4685), p-AKT (Lot: #4060), PI3K (Lot: #4257), caspase-3 (Lot: #9662), and cleaved caspase-3 (Lot: #9664, Cell Signaling Technology, Boston, MA, USA) overnight at 4°C . After rewarming for 30 min at room temperature, the membranes were washed and then incubated with horseradish peroxidase-conjugated secondary antibody (Lot: #7074/#7076) for 2 h. The blots were developed with an enhanced chemiluminescence (ECL) reagent (Millipore, Boston, MA, USA) and visualized using Image Lab Software (Bio-Rad, Hercules, CA, USA).

2.13. Histology and immunofluorescence staining

After fixing in 4% paraformaldehyde, brain and colon tissues were paraffin-embedded and cut into 4- μm -thick slices. For histology, the brain tissues were stained with hematoxylin and eosin (H&E) for analysis of morphometric lesions, while colon tissues were subjected to Alcian blue periodic acid–Schiff (AB-PAS) staining for the determination of the number of goblet cells. For TUNEL staining of the hippocampus, coronal slides of half-brain tissue were incubated with 50 μL of a TUNEL reaction mixture, with 50 μL of Label Solution serving as the negative control. For ZO-1 and OCLN double-immunofluorescence staining in colon tissue, slides were incubated with a rabbit antibody against ZO-1 (Lot: 21773-1-AP) and a mouse antibody against OCLN (Lot: 66378-1-Ig, Proteintech). After washing, the slides were incubated with Alexa Fluor 488-conjugated goat anti-mouse secondary antibody (Lot: A10680) and CY3-conjugated goat anti-rabbit secondary antibody (Lot: A10520, Thermo Fisher Scientific). Finally, all the slides were washed three more times with PBS and counterstained with DAPI.

2.14. Determination of SCFA contents

Colonic feces or brain tissues were freshly collected and homogenized in 3% phosphoric acid solution ($w/v = 1:10$). The supernatants of the feces, brain tissues, and *in vitro* bacterial cultures were obtained by centrifugation at 10,000 rpm for 10 min at 4°C . Then, *n*-butyl alcohol was added to the supernatants ($v/v = 1:1$) and SCFAs were extracted by vortexing. The upper *n*-butyl alcohol layer containing the SCFAs was collected and injected into gas chromatography–tandem mass spectrometry (GC–MS) for the determination of SCFA contents.

GC–MS analysis was performed in a GCMS-TQ8040NX system (Shimadzu, Japan) coupled to a high-polarity capillary column (wax-DA Cap Column, 0.25 mm $\times$ 0.25 μm $\times$ 30 m; Shimadzu). The injector, ion source, and interface temperatures were 240, 230, and 240°C , respectively. The flow rate of the helium carrier gas was kept at 1 mL/min 1 μL aliquot of each sample was automatically injected with a 3.5 min solvent delay and a 10:1 split ratio for fecal and bacterial culture samples or splitless injection for plasma and brain samples. The initial column temperature was 90°C and was held for 1 min, and then ramped up to 200°C at a rate of $10^\circ\text{C}/\text{min}$, and held at 200°C for 2 min. Ionization was performed in electron impact mode at 70 eV. The identification of SCFAs was confirmed by injection of pure standards and comparison of the retention time and the corresponding MS spectra. Analytes were detected in selected ion monitoring (SIM) mode using a target ion and reference ions and quantified using the external standard method. Data were analyzed using GCMSsolution Software (Shimadzu).

2.15. Statistical analysis

Data with a greater than two-fold standard deviation were discarded. All data were analyzed using GraphPad Prism 8.0 (GraphPad Software, San Diego, CA, USA) and are presented as means $\pm$ standard error of the mean (SEM). Unpaired *t*-tests were used for comparisons between two groups of continuous variables. One-way analysis of variance (ANOVA) followed by a post hoc test was employed for comparisons among three or more groups. Pearson's correlation analysis was used to assess the association between two continuous variables.

3. Results

3.1. The chemical profile of DZSM

The chemical profile of DZSM is shown in Fig. S1. A total of 24 compounds, including 5 phenolic acids, 4 flavonoids, 8 saponins, and 7 lignans, were quantified using UHPLC–HRMS. The contents of these compounds are shown in Table S2. The quantitation results revealed that all the ingredients had excellent linearity and that the accuracy and precision were within the recommended linear limits.

3.2. The gut microbiota mediated the ameliorative effect of DZSM on cognition-related disorder in 2VO model rats

As shown in Fig. 1A, in the orientation navigation test, the escape latency of rats in the model group was significantly longer than that of rats in the sham group; however, DZSM-treated animals had significantly shorter escape latency compared with rats in the model group. Representative paths of the rats in the different groups during the orientation navigation test are shown in Fig. 1B. A marked increase in the number of crossing the original platform location and the time spent

in the target quadrant was observed after treatment with DZSM during the spatial exploration test, along with a decrease in the time spent searching in the original platform location (Fig. 1C). These results indicated that DZSM exerted a significant ameliorative effect on cognitive impairment in 2VO model rats. However, the loss of gut microbiota through the administration of broad-spectrum antibiotics greatly weakened some of the beneficial effects of DZSM, suggesting that the gut microbiota may mediate the ameliorative effect of DZSM against cognitive dysfunction. The hippocampus performs important functions in both memory formation and learning. Thus, H&E staining and evaluation of oxidative stress in this organ were used to further confirm the beneficial effect of DZSM treatment on cerebral ischemia. The results showed that 2VO-induced neuropathological changes, including neuronal shrinkage and the occurrence of hyperchromatic nuclei, were significantly improved after treatment with DZSM (Fig. 1D). Moreover, 2VO rats displayed significantly higher levels of MDA and lower levels of SOD in plasma and hippocampal tissue relative to those of the sham group, whereas these abnormalities showed a notable recovery following DZSM intervention (Fig. 1E and F). However, like that seen in the behavior test, the protective effects of DZSM on hippocampal neurons and oxidative stress were substantially reduced after antibiotic application. These results revealed that the gut microbiota might play an important role in the DZSM treatment-mediated improvement in cognitive dysfunction in 2VO model rats.

3.3. DZSM modulated the gut microbiota and several key bacterial species

16S rRNA gene sequencing was performed to investigate the effects of DZSM on the regulation of the gut microbiota. The Shannon and Simpson index values indicated that the alpha-diversity of the gut microbiota was lower in rats with chronic cerebral ischemia than in those of the sham group. However, the alpha-diversity could be

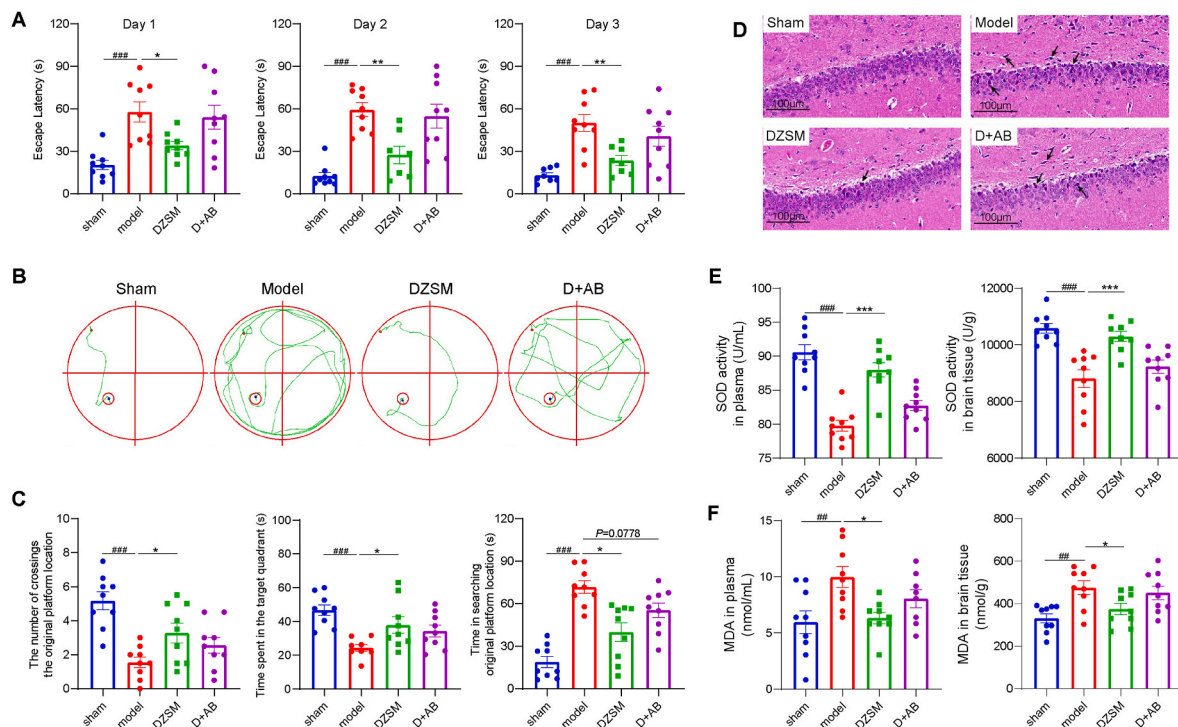


Fig. 1. Gut microbiota mediated the amelioration of DZSM on cognition-related disorders in 2VO model rats. (A) The escape latency during a 3-day training course in the orientation navigation test. (B) Representative swimming path of the SD rats in the orientation navigation test. (C) The number of crossings the original platform location, the time spent in the target quadrant, and the time in searching original platform location of rats in the spatial exploration test. (D) Representative images of H&E staining of the rat hippocampus (scale bars: 100 μ m). (E) SOD activity in plasma and hippocampus. (F) MDA activity in plasma and hippocampus. Data are shown as means $\pm$ SEM ($n = 8\text{--}10$). $^{\#}P < 0.05$, $^{\#\#}P < 0.01$, $^{\#\#\#}P < 0.001$ vs. the sham group. $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ vs. the model group; DZSM: dengzhan shengmai; SOD: superoxide dismutase; MDA: malondialdehyde.

markedly restored after the DZSM treatment (Fig. 2A). Principal coordinate analysis (PCoA) and partial least-squares discriminant analysis (PLS-DA) showed that cerebral ischemia substantially altered the overall structure of the gut microbiota compared with that of the sham group; nevertheless, DZSM treatment could reverse these shifts along PC1/COMP1 to some extent (Fig. 2B). These results indicated that DZSM partly restored the diversity and structure of the gut microflora in rats with chronic cerebral ischemia.

Phylum-level analysis identified a notable shift in the taxonomic distribution of the gut microbiota following chronic cerebral ischemia, as evidenced by the altered abundance of the two dominant phyla *Firmicutes* and *Bacteroidetes*. The 2VO model group had a higher abundance of *Firmicutes* and a lower abundance of *Bacteroidetes* compared with the sham group; however, DZSM treatment significantly restored the relative abundance of these two phyla (Fig. 2C and D). LEfSe analysis also indicated that *Bacteroidetes* was enriched in the DZSM group while *Firmicutes* was enriched in the model group (Fig. 2E). Microbiota community heatmap analysis at the family and genus levels showed that in rats with chronic cerebral ischemia, the relative abundance of several bacteria with anti-inflammatory properties was decreased, including those of members of the *Prevotellaceae* (e.g., *Prevotellaceae_NK3B31_group*, *Prevotellaceae_Ga6A1_group*, *Prevotellaceae_UCG-001*), *Bacteroidaceae*, and *Alloprevotella*. In contrast, the abundance of these beneficial bacteria was partly restored after DZSM intervention (Fig. 2F and G). The abundance of probiotic bacteria such as *Bacteroidaceae*, *Lactobacillus*, *Allobaculum*, *Ruminococcus torques*, and *Ruminococcus gnavreaii* was also greatly increased in the DZSM group according to the community heatmap analysis. Intriguingly, many of the microbiota displaying significantly elevated abundance with DZSM treatment, such as *Lactobacillus*, *Allobaculum*, *Alloprevotella*, *Prevotellaceae*, *Ruminococcus*, and *Bacteroidaceae*, are important SCFA-producing bacteria. These results revealed the beneficial regulatory effect of DZSM on the gut microbiota, characterized by the promotion of the growth of SCFA-producing bacteria. Additionally, the abundances of several opportunistic pathogens, including *Anaerostipes*, *Monoglobaceae*, *Monoglobus*, *Erysipelotrichaceae*, *Erysipelatoclostridium*, *Oscillospira*, *Lachnospiraceae_ND3007*, *Lachnospiraceae_UCG-004*, and *Candidatus Saccharimonas*, were decreased after DZSM treatment.

3.4. DZSM promoted the production and transportation of intestinal SCFAs

The effect of DZSM on the contents of nine main SCFAs in 2VO rat feces was assessed by GC-MS. We found that 4 weeks of DZSM treatment led to a significant increase in the levels of some intestinal SCFAs, including acetate, propionate, and butyrate (Fig. 3A and Fig. S2). However, the loss of gut microbiota resulted in a significant decline in the contents of these three SCFAs. To further evaluate how DZSM treatment affects SCFA production, the gut microbiota was isolated from the fresh feces of model rats and incubated *in vitro* under anaerobic conditions. As shown in Fig. 3B, DZSM treatment resulted in a dose-dependent increase in the production of the three important SCFAs; however, these effects were abolished following the heat-inactivation of the gut microbiota (Fig. S3), confirming that the facilitatory effect of DZSM on SCFA production was dependent on the gut microbiota. Furthermore, the mRNA expression of the key SCFA-producing enzymes acetate kinase (ACK), methylmalonyl-CoA decarboxylase (MMD), and butyryl-CoA: acetate-CoA transferase (BUT) in the feces of 2VO rats was markedly upregulated after DZSM treatment, providing further evidence for the promotive effect of DZSM on SCFA production (Fig. 3C). We also investigated the effects of DZSM ingredients on SCFA production at concentrations equivalent to those present in the high DZSM dosage (1000 µg/mL) and found that scutellarin, apigenin-7-O-glucuronide, ginsenoside Rg1/Rb1/Rg2, schisantherin A, and schisanrol A administration increased the levels of one or more SCFAs (Table 1).

Intestinal SCFAs must be transported into the circulation and brain

tissue to elicit their beneficial biological effects. The increase in SCFA levels in plasma and brain tissue following DZSM treatment suggested that DZSM enhanced the transportation of SCFAs (Fig. 3D). To confirm this possibility, the expression of two important monocarboxylic acid transporters (MCTs), MCT1 and SMCT1, was determined in the colons and brain tissues of 2VO rats, as well as in colonic Caco-2 cells. As shown in Fig. 3E and Fig. S4, treatment with DZSM and SCFAs resulted in a significant upregulation of the expression of MCT1 and SMCT1 in both colonic and cerebral tissues. However, these effects were lost with the removal of the gut microbiota, indicating that either the gut microbiota or their associated SCFAs played a crucial role in the DZSM-induced upregulation of these 2 transporters. However, we observed that the expression of the two MCTs was not increased in Caco-2 cells following DZSM intervention, whereas treatment with SCFAs, especially Na-PA and Na-BA, resulted in a significant increase in the expressions of both MCT1 and SMCT1 (Fig. 3F). These findings demonstrated that DZSM administration increased SCFA contents, which subsequently upregulated the expression of MCTs in colonic and cerebral tissue, thereby facilitating the transportation of SCFAs into the brain.

Furthermore, an analysis of Spearman's correlation between the abundance of gut microbiota and cognitive behaviors showed that the abundance of several important SCFA-producing bacteria, such as *Lactobacillus*, *Blautia*, *Phascolarctobacterium*, *Prevotellaceae*, and *Alloprevotella*, was significantly correlated with cognitive behaviors (Fig. S5). Meanwhile, a positive correlation was found between intestinal SCFA levels and learning and memory abilities (Fig. 3G). These results indicated that the restorative effects of DZSM on the cognitive and behavioral dysfunction after cerebral ischemia were mediated by the gut microbiota and associated SCFAs.

3.5. SCFAs mediated the anti-apoptotic effects of DZSM on cerebral neurocytes by activating the PI3K/AKT pathway

In this study, TUNEL staining of the rat hippocampus was used to analyze whether SCFAs were involved in the anti-apoptotic effects of DZSM. As shown in Fig. 4A, a greater number of apoptotic cells was observed in the model group than in the sham group. However, compared with the model group, DZSM treatment led to a significant decrease in the number of apoptotic cells, an effect that was weakened by antibiotic application. 2VO rats treated with SCFAs also exhibited significantly fewer apoptotic cells compared with the model group. We further evaluated the anti-apoptotic effect of DZSM and SCFAs in PC12 cells. The results showed that while DOX dose-dependently induced neuronal injury and apoptosis in PC12 cells (Fig. S6), treatment with SCFAs, but not DZSM, markedly abrogated these effects (Fig. 4B–D), suggesting that SCFAs might mediate the anti-apoptotic effect of DZSM.

Studies have reported that signaling via the PI3K/AKT/caspase-3 pathway represents an important mechanism underlying how SCFAs inhibit neuronal apoptosis (Wu et al., 2022; Zhou et al., 2021). Accordingly, we next analyzed the regulatory effect of DZSM or SCFA treatment on the PI3K/AKT/caspase-3 pathway in the rat hippocampus and PC12 cells. As shown in Fig. 4E, compared with the model group, treatment with DZSM and SCFAs led to a marked upregulation of PI3K and p-AKT levels and the downregulation of cleaved caspase-3 in the rat hippocampus; however, the promotive effect of DZSM on the expression of these proteins was weakened following antibiotic application. In PC12 cells, SCFA treatment significantly activated the PI3K/AKT pathway compared with that seen with DOX treatment, whereas DZSM treatment did not affect the protein expression of PI3K, p-AKT, or cleaved caspase-3 (Fig. 4F). These results demonstrated that SCFAs mediated the anti-apoptotic effect of DZSM on cerebral neurocytes via the PI3K/AKT/caspase-3 signaling pathway.

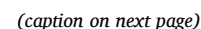


Fig. 2. DZSM modulated the gut microbiota and several key bacterial species. (A) Shannon and Simpson index of OTU level. (B) Microbiota community analysis based on PCoA and PLS-DA score plots. (C) Community abundance profiling of the gut microbiota on the phylum level. (D) The abundance (%) of *Firmicutes*, *Bacteroidetes*, and *Firmicutes/Bacteroidetes* ratio in the different groups. (E) Histograms of LDA scores for phyla with differential abundance identified using LEfSe analysis. (F) Community heatmap of the top 50 differentiated taxa on the family (upper) and genus (down) levels. (G) The mean proportion of multiple differentiated gut microbiota on the family (left) and genus (right) levels. Data are presented as means $\pm$ SEM ($n = 8-9$), * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. DZSM: dengzhan shengmai; OTU: operational taxonomic unit; PCoA: principal coordinate analysis; PLS-DA: partial least-squares discriminant analysis; LEfSe: linear discriminant analysis effect size.

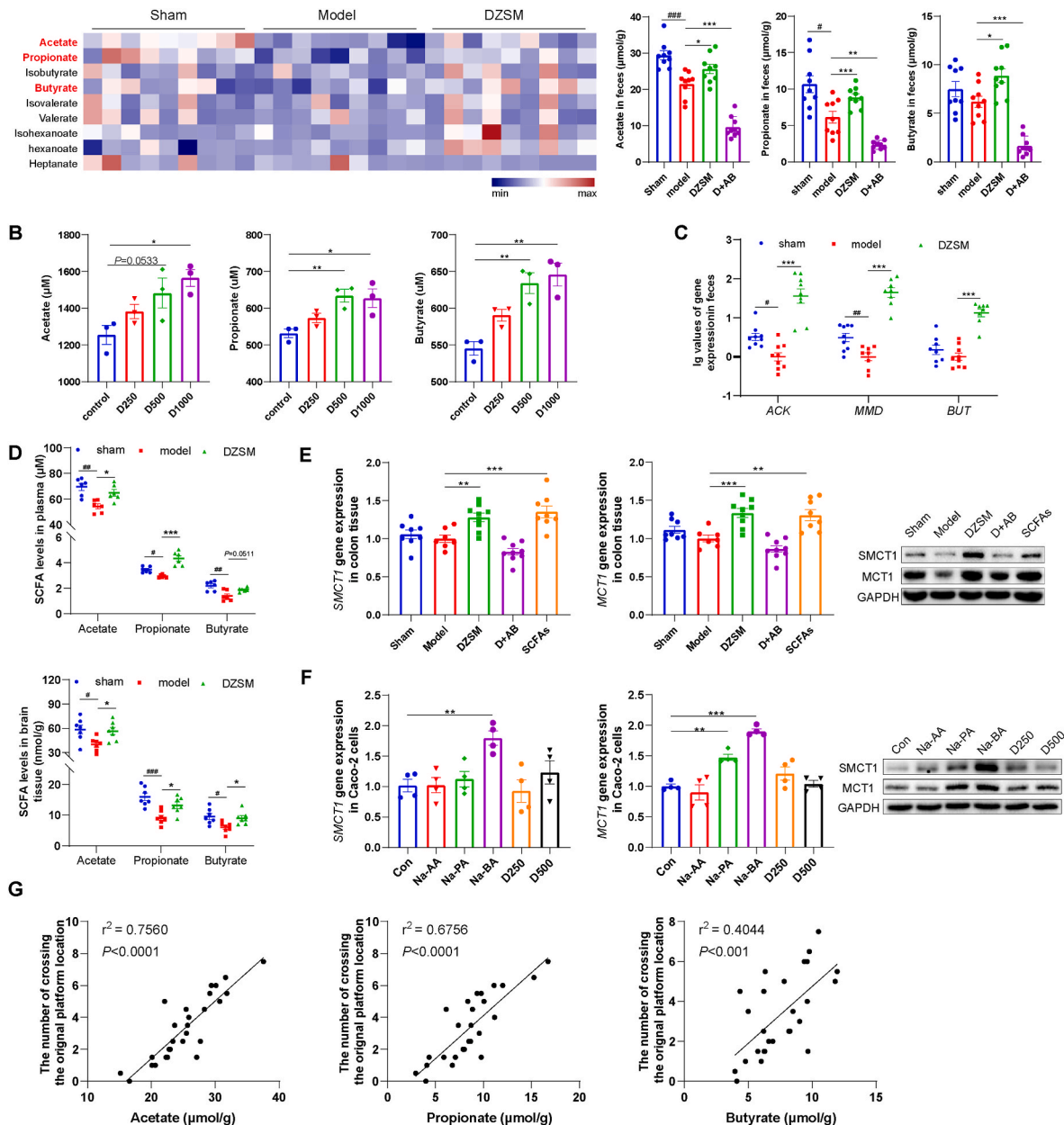


Fig. 3. Effect of DZSM treatment on the production and transport of fecal SCFAs. (A) SCFA concentrations in feces after 4 weeks of DZSM treatment in rats with 2VO-induced cerebral ischemia ($n = 9$). A-left: Heatmap analysis; A-right: Quantitative results of acetate, propionate and butyrate concentrations. (B) SCFA levels of the *in vitro* intestinal bacterial cultures derived from the feces of 2VO-treated SD rats after 18 h of DZSM treatment ($n = 3$). (C) The expression levels of the genes encoding the SCFA-producing enzymes ACK, MMD, and BUT in 2VO rat feces as determined by RT-qPCR ($n = 8-9$). (D) The mRNA expression and protein expression of the MCTs in colonic tissue of 2VO rats ($n = 7-9$). (E) The mRNA expression and protein expression of the MCTs in Caco-2 cells after SCFAs or DZSM intervention ($n = 4$). (F) SCFA levels of 2VO rat plasma after 4 weeks of DZSM treatment ($n = 6$). (G) Linear correlation analysis between SCFA levels and the number of crossings the original platform location in the spatial exploration test. Data are shown as means $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the rat model group or vs. the control group in intestinal bacteria cultures or Caco-2 cells; # $P < 0.05$, ## $P < 0.01$ vs. the sham group. DZSM: dengzhan shengmai; SCFA: short-chain fatty acid; 2VO: permanent occlusion of bilateral common carotid arteries; SD: Sprague-Dawley; ACK: acetate kinase; MMD: methylmalonyl-CoA decarboxylase; BUT: and butyryl-CoA: acetate-CoA transferase; MCT: monocarboxylate transporters.

Table 1SCFA levels of the *in vitro* intestinal bacterial cultures derived from the feces of 2VO-treated SD rats after 18 h of various DZSM ingredients treatment ($n = 3$).

Herbal origins	Compounds	Dose (μM)	Acetate (μM)	Propionate (μM)	Butyrate (μM)
<i>Erigeron breviscapus</i>	Control	/	1000.8 $\pm$ 36.1	492.7 $\pm$ 11.7	632.5 $\pm$ 19.9
	Scutellarin	250	963.7 $\pm$ 50.9	577.4 $\pm$ 29.1**	757.5 $\pm$ 20.4**
	Apigenin-7-O-glucuronide	50	1004.4 $\pm$ 79.1	524.3 $\pm$ 30.3	707.9 $\pm$ 27.3*
	4,5-dicaffeoylquinic acid	25	986.5 $\pm$ 37.6	487.9 $\pm$ 33.4	649.9 $\pm$ 34.9
	3-caffeoylquinic acid	50	1042.9 $\pm$ 71.4	537.4 $\pm$ 38.4	662.6 $\pm$ 22.3
	Caffeic acid	50	1069.8 $\pm$ 58.7	483.0 $\pm$ 10.5	632.7 $\pm$ 70.8
	3,4-dicaffeoylquinic acid	5	1018.9 $\pm$ 130.5	480.6 $\pm$ 49.6	675.0 $\pm$ 55.3
	3,5-caffeoylquinic acid	10	1054.6 $\pm$ 40.6	492.3 $\pm$ 23.5	652.8 $\pm$ 48.4
<i>Panax ginseng</i>	Ginsenoside Rg1	10	1185.4 $\pm$ 67.0*	512.8 $\pm$ 46.5	730.3 $\pm$ 34.0*
	Ginsenoside Rb1	10	1052.0 $\pm$ 29.7	597.0 $\pm$ 43.1*	712.8 $\pm$ 30.9*
	Ginsenoside Rg2	5	1075.5 $\pm$ 59.0	511.9 $\pm$ 45.4	735.1 $\pm$ 40.5*
	Ginsenoside Rc	2	1056.6 $\pm$ 40.1	522.4 $\pm$ 36.4	653.8 $\pm$ 28.1
	Ginsenoside Rb2	2	1021.2 $\pm$ 101.3	489.4 $\pm$ 41.0	658.9 $\pm$ 37.2
	Ginsenoside Rd	2	1058.0 $\pm$ 54.4	525.0 $\pm$ 41.6	641.6 $\pm$ 25.5
<i>Schisandra chinensis</i>	schisandrol A	20	1208.8 $\pm$ 20.1***	567.3 $\pm$ 32.6*	800.6 $\pm$ 45.1*
	Schisantherin A	10	1082.1 $\pm$ 56.4	565.6 $\pm$ 37.7*	678.2 $\pm$ 19.8
	Schisantherin B	5	1028.9 $\pm$ 70.2	482.5 $\pm$ 11.5	652.0 $\pm$ 45.8
	Schizandrin C	2	1061.3 $\pm$ 52.5	488.1 $\pm$ 40.8	676.4 $\pm$ 27.9
	Schizandrin B	2	957.4 $\pm$ 32.2	514.5 $\pm$ 44.9	682.9 $\pm$ 38.4
	Schizandrin A	2	981.5 $\pm$ 57.0	513.4 $\pm$ 15.5	623.1 $\pm$ 39.5
	Gomisin D	2	1017.2 $\pm$ 48.0	496.4 $\pm$ 42.9	653.3 $\pm$ 34.7
<i>Ophiopogon japonicus</i>	Methylphosphogonone A	1	1010.1 $\pm$ 9.9	532.1 $\pm$ 38.1	641.9 $\pm$ 48.2
	Methylphosphogonanone A	1	1015.1 $\pm$ 61.7	509.3 $\pm$ 47.1	668.8 $\pm$ 37.5

Data are presented as Mean $\pm$ SD ($n = 3$). Significant differences were evaluated by unpaired *t* tests. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the control group.

3.6. SCFAs mediated the anti-neuroinflammatory effects of DZSM by improving intestinal barrier integrity

As shown in Fig. 5A, 2VO surgery led to severe neuroinflammation, as evidenced by an increase in the levels of pro-inflammatory factors in the brain, including IL-1 β , IL-6, and TNF- α . Both DZSM and SCFA significantly reduced the levels of these key pro-inflammatory factors relative to that observed in the model group. However, this effect of DZSM was markedly weakened by antibiotic application. Spearman's correlation analysis between the levels of pro-inflammatory factors and bacterial taxa showed that the abundance of several SCFA-producing bacteria, such as *Lactobacillus*, *Allobaculum*, *Alloprevotella*, *Prevotellaceae*, *Ruminococcus*, *Blautia*, and *Phascolarctobacterium*, displayed a significant negative correlation with the cerebral levels of IL-1 β , IL-6, and TNF- α (Fig. 5B). Moreover, linear correlation analysis indicated that intestinal SCFA levels were negatively correlated with cerebral inflammation (Fig. S7). However, none of the three SCFAs (acetate, propionate, and butyrate) inhibited the LPS-induced inflammatory response in PC12 cells at the concentrations tested (Fig. S8), which led us to explore how SCFAs mediated the anti-neuroinflammatory effects of DZSM.

When the intestinal barrier is impaired, the endotoxin LPS can translocate from the intestine to the brain, where it induces neuroinflammation, while gut microbiota-derived SCFAs have been reported to play a key role in the maintenance of the gut barrier (Kim et al., 2018). Thus, in the present study, we evaluated the integrity of the intestinal barrier in 2VO rats by determining the number of mucin-producing goblet cells and the expression levels of tight junction proteins in colonic tissues. As shown in Fig. 5C–E and Fig. S9, the number of mucin-producing goblet cells and the expression of two tight junction proteins, ZO-1 and OCLN, were significantly reduced following 2VO surgery, suggesting that gut barrier integrity was disrupted in rats with chronic cerebral ischemia. Treatment with DZSM or SCFA markedly reversed these defects compared with the model group, while no significant change was observed in the D + AB group. In Caco-2 cells, treatment with Na-PA and Na-BA, but not DZSM, resulted in a notable upregulation of ZO-1 and OCLN expression levels (Fig. 5F). These results suggested that the increase in SCFA levels contributed to the protective effect of DZSM on the gut barrier. Meanwhile, the plasma LPS level was

significantly elevated in the model group, likely due to intestinal barrier impairment, an effect that was mitigated following DZSM and SCFA intervention (Fig. 5G). However, antibiotics application weakened the beneficial effect of DZSM on intestinal barrier integrity. A significant negative correlation was also found between intestinal SCFA levels and plasma LPS concentrations (Fig. 5H). These results revealed that the SCFA-mediated improvement in gut barrier integrity represents a key mechanism underlying how DZSM exerts its ameliorative effects on neuroinflammation following cerebral ischemia. Moreover, the results of plasma fatty acid binding protein 2 (FABP2) and zonulin levels, two important biomarkers of gut epithelial tight junction barrier integrity (Stevens et al., 2018), further suggested that SCFAs mediated the anti-neuroinflammatory effect of DZSM by improving intestinal barrier integrity (Fig. S10).

3.7. The effects of DZSM on ameliorating cerebral ischemia could be recapitulated by FMT

To further demonstrate the role of gut microbiota-derived SCFAs in the anti-cerebral ischemia effect of DZSM, fresh fecal bacteria derived from model (model-FMT) or DZSM-treated (DZSM-FMT) donor rats were administered to 2VO rats by gavage. PCA and PCoA analysis revealed that the model-FMT and DZSM-FMT groups were segregated into different clusters (Fig. 6A–a). Analysis of the taxonomic composition of the recipient gut microbiota at the phylum level showed that gavage with DZSM-treated rat feces increased the relative abundance of *Bacteroidetes* while reducing that of *Firmicutes* when compared with the administration of fecal bacteria derived from model animals (Fig. 6A and b), which was consistent with that observed in the donor rats. At the genus level, the abundance of multiple SCFA-producing bacteria, including *Allobaculum*, *Alloprevotella*, *Bacteroides*, *Akkermansia*, *Ruminiclostridium*, *Butyrivibrio*, and *Lactobacillus*, was significantly greater in the DZSM-FMT group than in the model-FMT group (Fig. 6A–c). Moreover, the levels of fecal SCFAs and the expression of the MCT1 and SMCT1 genes in the colon were found to be upregulated with the administration of fecal bacteria from DZSM-treated donor rats (Fig. 6B and C). This suggested that FMT could reproduce the promotive effects of DZSM on SCFA production. Furthermore, gut barrier integrity was improved in rats of the DZSM-FMT group compared with that in rats of

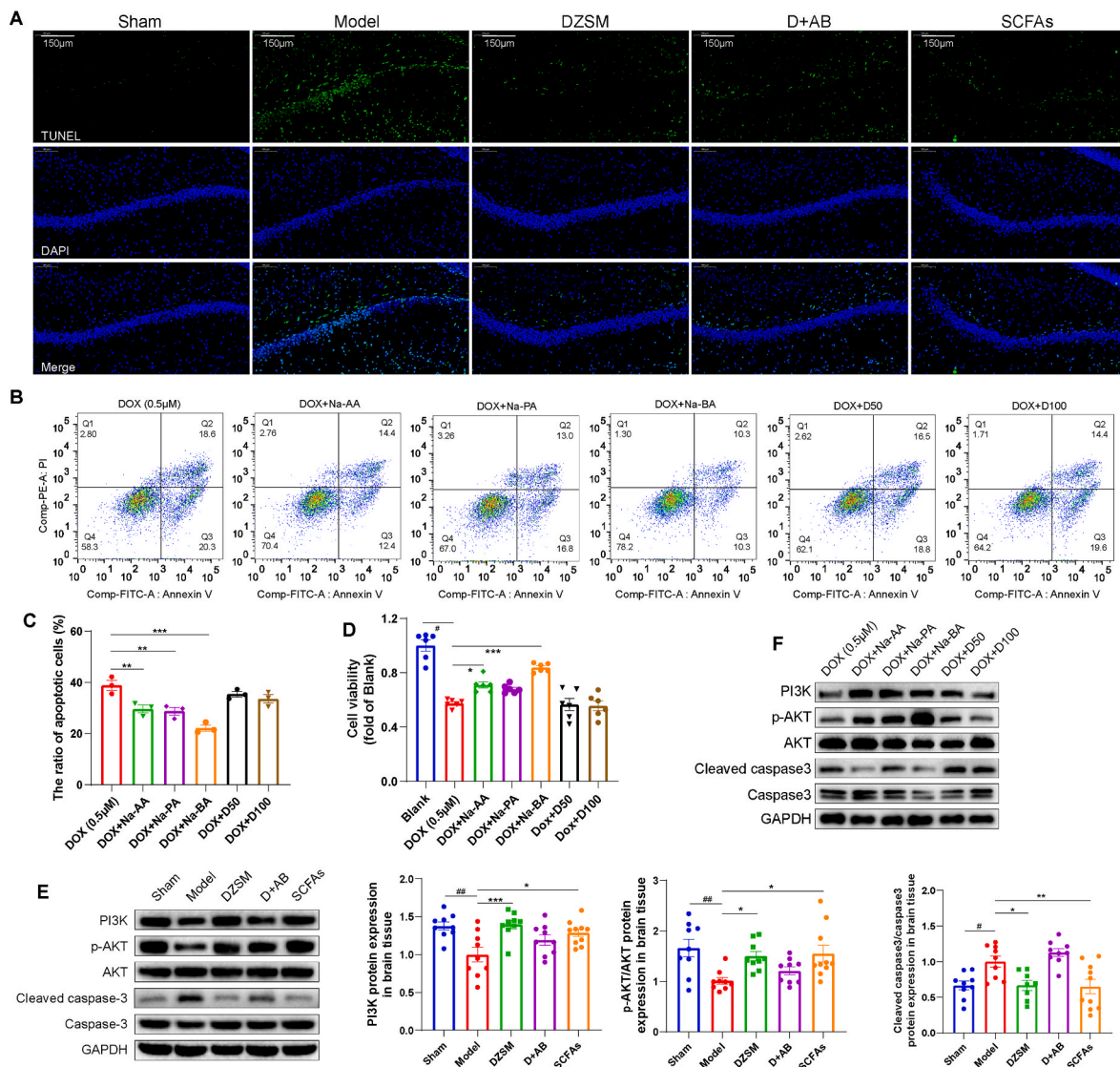


Fig. 4. SCFAs mediated the anti-apoptotic effects of DZSM on cerebral neurocytes by activating PI3K/AKT pathway. (A) Representative images of TUNEL staining of the rat hippocampus (scale bars: 150 μ m). (B, C) Representative plots (B) and quantitative results (C) of apoptosis detection by flow cytometry using Annexin V-FITC/PI staining in DOX (0.5 μ M)-treated PC12 cells after drug treatment ($n = 3$). (D) Analysis of cell viability in DOX (0.5 μ M)-treated PC12 cells after drug treatment ($n = 6$). (E) Protein expression of PI3K/AKT/caspase-3 (apoptotic signaling pathway) in the hippocampus of 2VO rats ($n = 8\sim 10$). (F) Protein expression of PI3K/AKT/caspase-3 in DOX-treated PC12 cells after drug treatment. Data are shown as means $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the model group or the DOX (0.5 μ M) group; # $P < 0.05$ vs. the sham group or the untreated (blank) group. DZSM: dengzhan shengmai; SCFA: short-chain fatty acid; DOX: doxorubicin.

the model-FMT group, as evidenced by the levels of plasma LPS and colonic epithelial tight junction proteins ZO-1 and OCLN expression (Fig. 6D–F). Accordingly, cognition dysfunction and neuronal apoptosis were significantly improved in DZSM-FMT rats compared with that observed in the model-FMT rats (Fig. 6G and H).

4. Discussion

Increasing evidence has indicated that the gut microbiota plays an important role in various diseases via its interaction with other organs, including the gut–liver axis, the gut–brain axis, the gut–heart axis, and even the gut–lung/kidney axis (Albillos et al., 2020; Dang and Marsland, 2019; Dinan and Cryan, 2017; Sumida and Kovacs, 2019). Among them, the gut–brain axis represents a crucial link mediating the relationship between the gut microbiota and neurological conditions, including cerebral ischemic diseases (Pluta et al., 2021). Targeting the gut microbiota–gut–brain axis may be a potential therapeutic option for

cerebral ischemia. In this study, we elucidated the effect of DZSM in ameliorating cerebral ischemia from the perspective of the gut microbiota for the first time and revealed that DZSM effectively modulates the composition of the gut microbiota and the production of bacteria-derived SCFAs. The increased transportation of intestinal SCFAs into cerebral tissue effectively inhibited the apoptosis of neurocytes. Moreover, the increase in intestinal SCFA levels was found to attenuate neuroinflammation by promoting the repair of the damaged gut barrier following 2VO surgery. Consequently, cognitive dysfunction, cerebral neuropathy, and oxidative stress were all improved, leading to the amelioration of chronic cerebral ischemia (Fig. 7).

DZSM, a well-known TCM formula, is widely used in secondary prevention after ischemic stroke and ranks first among TCMs for the treatment of neurological conditions (Mu et al., 2019). DZSM boosts immunity by benefiting ‘Qi’ and nourishing ‘Yin’, invigorating blood circulation, and strengthening the brain based on traditional Chinese medicine theory (Yang et al., 2018). Pharmacology-based studies over

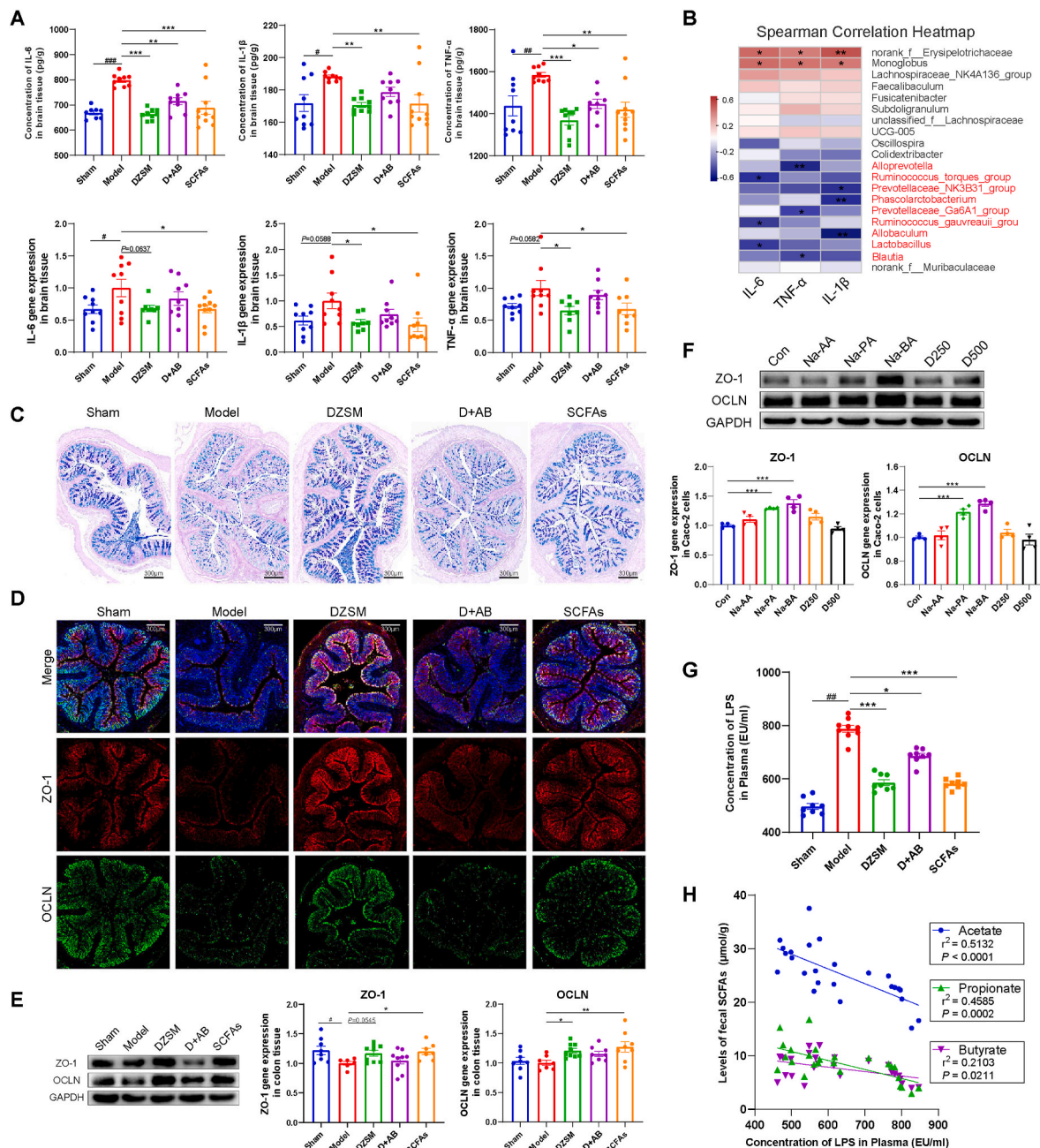


Fig. 5. SCFAs mediated the anti-neuroinflammation of DZSM through improving intestinal barrier. (A) The levels of IL-6, IL-1 β , and TNF- α in brain tissue of SD rats as determined using ELISA (up) and RT-qPCR (down) ($n = 8\sim 10$). (B) Spearman's correlation heatmap analysis between the gut microbiota and cerebral pro-inflammatory factors. (C) Representative images of colonic goblet cells using AB-PAS staining in SD rats (scale bars: 300 μ m). (D) Representative images of immunofluorescence staining for ZO-1 (red) and OCLN (green) in colonic tissue of SD rats (scale bars: 300 μ m). (E) The expression of the tight junction proteins ZO-1 and OCLN in SD rats as determined by western blotting (left) and RT-qPCR (right) ($n = 7\sim 10$). (F) The expression of the tight junction proteins ZO-1 and OCLN in Caco-2 cells as determined by western blotting (up) and RT-qPCR (down) ($n = 4$). (G) Plasma LPS levels in SD rats as determined by ELISA ($n = 8\sim 10$). (H) Linear correlation analysis between fecal SCFA levels and plasma LPS levels. Data are shown as means $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the rat model group or the control group in Caco-2 cells; # $P < 0.05$, ## $P < 0.01$, ### $P < 0.001$ vs. the sham group. DZSM: dengzhan shengmai; SCFA: short-chain fatty acid; SD: Sprague–Dawley; LPS: lipopolysaccharide. AB-PAS: alcian blue periodic acid–Schiff. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the last decade have indicated that the active chemical constituents of DZSM mainly include compounds such as flavonoids, phenolic acids, lignans, and saponins. Among them, scutellarin, a flavonoid derived from *Erigeron breviscapus*, is considered to be the main active ingredient in the treatment of chronic cerebral ischemia and exerts its effects by regulating glutamatergic and GABAergic synaptic inputs, the MAPK signaling pathway, or microglial and astrocytic activities (Chen et al., 2020; Fang et al., 2016; Sheng et al., 2020). 4,5-Dicaffeoylquinic acid

and 3,5-dicaffeoylquinic acid, phenolic acids obtained from *Erigeron breviscapus*, both exert therapeutic effects on cerebral ischemia-induced neuronal injury in rats (Liu et al., 2021). Ginsenosides, especially Rb1, protect against cerebral ischemic injury by inhibiting excitotoxicity and Ca^{2+} influx, preserving blood–brain barrier (BBB) integrity, and maintaining energy metabolism (Xie et al., 2021). Schizandrin A, a lignan isolated from *Schisandra chinensis*, can protect against cerebral ischemia–reperfusion injury by regulating the AMPK/Nrf2 pathway (Zhou

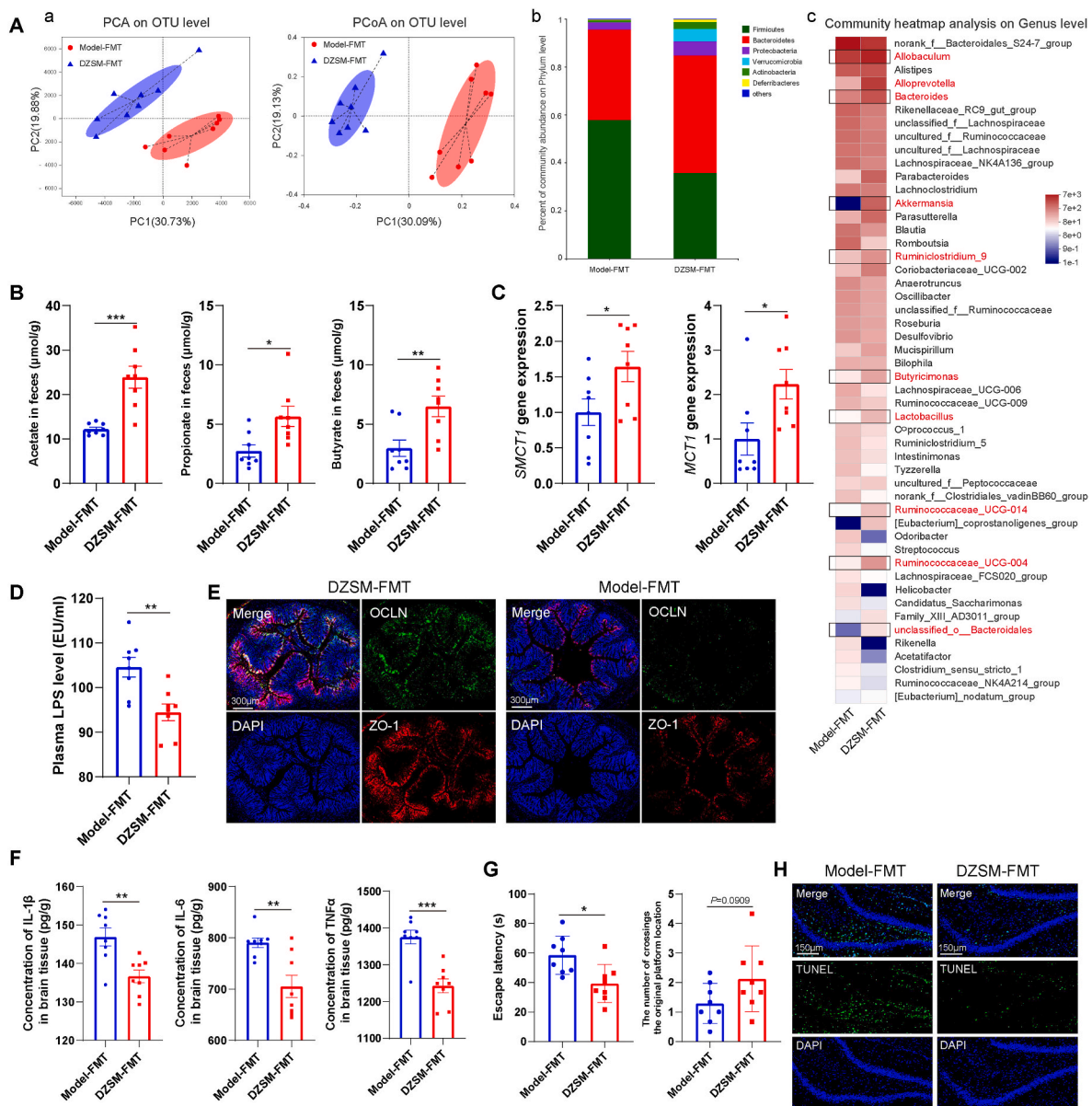


Fig. 6. The effects of DZSM on ameliorating cerebral ischemia can be reproduced by FMT. Fresh fecal bacteria from the model group or DZSM group donor rats were transplanted to the antibiotics-treated 2VO recipient rats (group: Model-FMT, DZSM-FMT) by oral gavage for 4 weeks ($n = 8$). (A) Microbiota community analysis of recipient rats. A-a: PCA and PCoA score plots; A-b: Community abundance profiling of the gut microbiota at the phylum level. A-c: Heatmap of the top 50 differentiated taxa on the genus level. (B) The levels of fecal SCFAs in recipient rats. (C) The mRNA expression of the MCTs genes in colonic tissue of the recipient rats. (D) The plasma LPS level of the recipient rats. (E) Representative images of immunofluorescence staining for OCLN (green) and ZO-1 (red) in colonic tissue of the recipient rats (scale bars: 300 μm). (F) The levels of IL-6, IL-1 β , and TNF- α in brain tissue of recipient rats. (G) The escape latency of the orientation navigation test (left) and the number of crossings the original platform location of the spatial exploration test (right). (H) Representative images of TUNEL staining of the recipient rat hippocampus (scale bars: 150 μm). Data are shown as means $\pm$ SEM. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ vs. the Model-FMT group. FMT: Fecal microbiota transplantation; DZSM: dengzhan shengmai; PCA: Principal component analysis; PCoA: principal coordinate analysis; SCFA: short-chain fatty acid. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

et al., 2019). Despite the importance of these studies relating to DZSM ingredients, none have assessed the importance of gut microbiota in mediating the effects of DZSM. Indeed, many important active components of DZSM have very poor bioactivity *in vivo* and aggregate at relatively high concentrations in the intestinal tract. For example, scutellarin, a BCS class IV drug, has very low bioavailability due to its low solubility in water and poor BBB-permeating ability (Dang et al., 2019). Multidrug resistance-associated protein 2-induced efflux in the intestine further aggravates the poor absorption and oral bioavailability of scutellarin (Xiao et al., 2016). Other active ingredients, such as phenolic

acids, lignans, and saponins, either are present at extremely low concentrations in DZSM or have very poor solubility and absorption, thus reducing their pharmacological effects *in vivo* compared with that observed *in vitro*. However, all of the unabsorbed ingredients of DZSM, irrespective of their properties, interact with the gut microbiota and intestinal epithelial cells. In the present study, the application of antibiotics in 2VO rats weakened the effects of DZSM, indicating that the gut microbiota contributed to the anti-cerebral ischemic effect of this TCM formula. In addition, the results of fecal 16S rRNA gene sequencing demonstrated that DZSM significantly modulated the diversity,

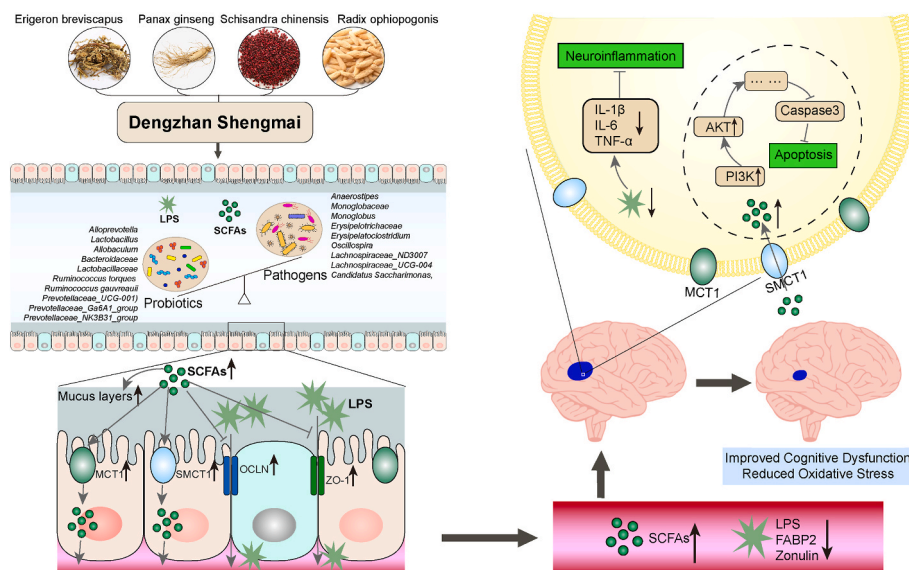


Fig. 7. Schematic illustration showing the mechanism of DZSM in ameliorating cerebral ischemia via microbiota-gut-brain axis. LPS: lipopolysaccharide; SCFA: short-chain fatty acid; SMCT1: sodium-coupled monocarboxylate transporter 1; MCT1: monocarboxylate transporter 1; FABP2: fatty acid binding protein 2.

structure, and composition of the gut microbiota. Furthermore, FMT could reproduce the effects of DZSM in cerebral ischemia, confirming that the anti-cerebral ischemic effects of DZSM are at least partially mediated by the gut microbiota.

The gut microbiome is a complicated ecosystem hosting trillions of microbes that associate with the host in a commensal or opportunistic manner. It is dominated by two bacterial phyla—Firmicutes and Bacteroidetes—that together account for more than 90% of the total gut microbiome. Therefore, fluctuations in the abundance of either phylum can reflect overall changes in bacterial composition. Alterations in the abundance of these two bacteria have been associated with several pathological conditions, including cerebral ischemic stroke (Pluta et al., 2021; Socala et al., 2021). In the present study, we found that DZSM restored the relative abundance of Bacteroidetes and Firmicutes, which had been disrupted after 2VO surgery, revealing that DZSM exerted a beneficial regulatory role in the modulation of the gut microbiota. SCFAs, the main metabolites produced by bacterial fermentation of dietary fiber and resistant starch in the colon, are speculated to play a key role in microbiota-gut-brain axis communication and the treatment of cerebral ischemic stroke through multiple mechanisms (Dalile et al., 2019; Yang, L.L. et al., 2020b). *Lactobacillus*, an important SCFA-producing bacterium, was reported to be significantly reduced after infarction of the brain, and its supplementation was shown to improve cognitive function and mood, as well as alleviate neuroinflammation (Hwang et al., 2019). *Allobaculum*, *Ruminococcus*, *Prevotella*, and *Alloprevotella* are also important SCFA-producing bacteria and have been reported to be negatively correlated with inflammation (Kong et al., 2019; Shang et al., 2018). In the present study, DZSM administration led to a marked increase in the abundance of these SCFA-producing bacteria. Meanwhile, the analysis of fecal SCFA levels by GC-MS demonstrated that DZSM treatment enhanced the production of gut microbiota-derived SCFAs. We also found that the abundance of multiple SCFA-producing bacteria and intestinal SCFA levels were positively correlated with cognitive function. These results not only confirmed that SCFAs exert a protective effect against cerebral ischemic diseases, but also suggested that gut microbiota-derived SCFAs mediate the therapeutic effects of DZSM. Following their production in the colon, SCFAs must be absorbed by enterocytes and then enter the blood circulation and brain to exert their biological effects (Dalile et al., 2019; Silva et al., 2020). In the present study, we found that the DZSM-induced increase in intestinal SCFA levels resulted in significant upregulation of

MCT expression in enterocytes and neurocytes, thus leading to an increase in cerebral SCFA levels.

Neuronal apoptosis is considered the main cause of neuronal death following cerebral ischemia (He et al., 2021; Tang et al., 2020), and inhibiting neuronal apoptosis is especially important for attenuating neurological deficits. An increase in the levels of SCFAs in the brain, especially butyrate, was shown to attenuate neuronal apoptosis and the expression of oxidative stress markers via activating the PI3K/AKT pathway (Li et al., 2018; Zhou et al., 2021). Propionate has also been reported to possess anti-apoptotic potential through the inhibition of histone deacetylase activity, the promotion of NF- κ B signaling, or the activation of the PI3K/AKT signaling pathway (Ali et al., 2021; Wu et al., 2022). In this study, 2VO-induced apoptosis of cerebral neurocytes was reversed by DZSM and SCFA intervention, whereas the loss of gut microbiota significantly weakened the effects of DZSM. Subsequently, employing PC12 cells, we demonstrated that the anti-apoptotic effects of DZSM in cerebral neurocytes were mediated by gut microbiota-derived SCFAs. Notably, this was also the first study to show that acetate can inhibit cerebral ischemia-induced neuronal apoptosis by regulating the PI3K/AKT pathway.

The inhibition of neuroinflammation also promotes the improvement of cerebral ischemic injury. Studies have shown that DZSM and its active ingredients, including scutellarin, ginsenoside Rb1/Rg1, and schizandrin, can reduce neuroinflammation (Bian et al., 2020; Wang et al., 2021; Yan et al., 2021). In this study, the results of DZSM-antibiotics co-administration indicated that either the gut microbiota or intestinal SCFAs were involved in the DZSM-modulated inhibition of neuroinflammation. Correlation analysis between cerebral pro-inflammatory factors and the gut microbiota or intestinal SCFA levels further revealed that the increased SCFA levels were correlated with the amelioration of neuroinflammation (Wenzel et al., 2020). reported that SCFAs reduce the inflammatory response in THP-1 microglia-like cells by reducing cytotoxin levels. However, in the present study, we observed that SCFA treatment did not directly influence LPS-induced inflammation in PC12 cells. We think that the use of different types of cells, different SCFA concentrations and combinations, and different inflammatory inducers led to the different outcomes. It is feasible that SCFA may also directly inhibit LPS-induced inflammation in PC12 cells if administered at higher concentrations (>100 μ M); however, it is impossible to reach such high SCFA concentrations in the brain and was thus not relevant to the pharmacological mechanism of

DZSM. Accordingly, we next sought to determine how intestinal SCFAs mediated the ameliorative effects of DZSM against neuroinflammation. Intestinal barrier dysfunction results in the translocation of enteric pathogens and endotoxins, followed by the aggravation of inflammation and neurocyte necrosis (Wei et al., 2021). Meanwhile, an increase in SCFA levels can restore intestinal barrier function by promoting the production of mucus, selectively upregulating the expression of tight junction proteins, and enhancing tight junction assembly (Xiao et al., 2020). The results of the present study revealed that intestinal SCFAs significantly improved the integrity of the gut barrier in 2VO rats, which largely explains why DZSM could ameliorate cerebral ischemia-induced neuroinflammation.

In this study, we focused on the effects of DZSM in ameliorating cerebral ischemia via the regulation of the gut microbiota and SCFA production based on the interaction of unabsorbed components of DZSM with the gut microbiota; however, it is important to also consider the contribution of the absorbed DZSM ingredients. Although the application of broad-spectrum antibiotics could effectively eliminate the intestinal microbiota-mediated effects of DZSM on cognitive function, it cannot be excluded that antibiotic administration may have affected the experimental results by directly interacting with DZSM components. Here, we administered antibiotics and DZSM at different time points to reduce the drug–drug interaction; nevertheless, drug-induced changes in P-gp and CYP450 expression could not be eliminated when the drugs were continuously administered over a long period. Accordingly, our results relating to DZSM-antibiotics co-administration (D + AB group) in the 2VO rats cannot fully represent the efficacy of the absorbable constituents of DZSM. Future studies should consider the use of germ-free animals to avoid these limitations.

5. Conclusions

In this study, we demonstrated that the therapeutic effects of DZSM on cerebral ischemia are at least partially mediated by gut microbiota-derived SCFAs. DZSM restored gut microbiota homeostasis, promoted the growth of SCFA-producing bacteria, and elevated intestinal SCFA levels, which improved gut barrier integrity and reduced neuroinflammation after 2VO surgery. We further showed that DZSM treatment increased the transportation of intestinal SCFAs into cerebral tissue, which effectively inhibited neurocyte apoptosis. Consequently, DZSM alleviated cerebral ischemia-induced cognitive dysfunction. Our findings may contribute to our understanding of why TCM exhibits good efficacy despite the poor absorption of many of its key ingredients.

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CRedit authorship contribution statement

Hui-Hui Guo: performed the experiments, analyzed the data and drafted the manuscript. **Hao-Ran Shen:** performed the experiments and analyzed the data. **Ming-Ze Tang:** performed the animal experiments. **Ning Sheng:** provided the chemical profiles of DZSM. **Xiao Ding:** critically revised the manuscript for key intellectual content. **Yuan Lin:** critically revised the manuscript for key intellectual content. **Jin-Lan Zhang:** critically revised the manuscript for key intellectual content. **Jian-Dong Jiang:** designed the study and modified the manuscript. **Tian-Le Gao:** analyzed the data, directed the project and revised the

manuscript. **Lu-Lu Wang:** directed the project and modified the manuscript. **Yan-Xing Han:** designed the study and approved the final version of the manuscript which to be submitted.

Declaration of competing interest

All the authors had no potential conflict of interest to declare.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jep.2023.116158>.

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